



COMPLETE LOGICAL REASONING

CHEAT SHEET

All 20 Topics | Beginner → Advanced | Placements | CAT | UPSC

Learn How to Think, Not Just Solve

1 CODING-DECODING

A. What is it?

In this topic, you're given a secret code/pattern. You need to figure out the RULE and apply it to new situations. Think of it like finding the hidden pattern in a treasure map.

B. Core Concepts

- **Alphabet Coding:** Letters are replaced by other letters ($A \rightarrow B$, $B \rightarrow C$, etc.)
- **Number Coding:** Letters become numbers ($A=1$, $B=2$, etc.)
- **Position Coding:** Based on position in alphabet (Reverse: $Z=1$, $Y=2$, etc.)
- **Mathematical Coding:** Apply math operations to letter positions
- **Substitution Coding:** Fixed mapping ($A \rightarrow X$, $B \rightarrow Y$, etc.)

C. Types of Questions

- Code one word when given the pattern
- Decode a word given its coded form
- Find the pattern by comparing coded examples
- Apply pattern to find code for multiple words

D. Step-by-Step Solving Approach

1. **Identify the type:** Is it alphabet, number, or pattern-based?
2. **Find the rule:** Compare 1-2 given examples to find pattern
3. **Test the rule:** Verify with second example if available

4. **Apply consistently:** Use the rule on the question word
5. **Double-check:** Make sure your answer follows the pattern

E. Shortcuts & Tricks

- 💡 **Quick Pattern Finding:** Always check if it's +1, +2, or reverse pattern first (most common)
- 💡 **Alphabet Position Method:** A=1, B=2, Z=26. If code is +3 shift, A becomes D
- 💡 **Reverse Alphabet:** Z=1, Y=2, A=26. Check this if +1 shift doesn't work
- 💡 **Pattern in Whole Word:** Sometimes each letter has same rule, sometimes different letters have different rules

F. Solved Examples

BASIC: If PLANE is coded as QMBLF, how is TRAIN coded?

Solution:

P→Q (+1), L→M (+1), A→B (+1), N→O (+1), E→F (+1)

Rule: Each letter shifts by +1

Apply to TRAIN: T→U, R→S, A→B, I→J, N→O

Answer: USBJO

INTERMEDIATE: If MOTHER is coded as RGVSBQ, how is SISTER coded?

Solution:

M(13)→R(18) (+5), O(15)→G(7) (-8)... Finding pattern is confusing.

Try another: M→R is +5, O→G is -8? No clear pattern.

Check position-based: 1st letter +5, 2nd letter -8, 3rd letter +5...

Actually: M→R(+5), O→G(-8), T→V(+2), H→S(+11), E→B(-3), R→Q(-3)

This might be reverse mapping. Check: Position in MOTHER

Found: Each position has its own rule. This requires testing with given answer.

ADVANCED: In a certain code, all vowels become next consonant and consonants become previous letter. CAT is coded how?

Solution:

C (consonant) → B (previous letter)

A (vowel) → B (next consonant from A)

T (consonant) → S (previous letter)

Answer: BBS

G. Common Mistakes

- ✗ Not testing the pattern with all given examples
- ✗ Assuming same rule for all letters (sometimes it's position-dependent)
- ✗ Forgetting that rule can be reverse alphabet, not just forward

H. Exam Strategy

- PLACEMENTS: Pattern is usually simple (+1 or +2 shift). Time: 1-2 min per question
- CAT: Pattern is complex, needs careful analysis. Time: 3-5 min per question
- UPSC: Usually straightforward. Focus on accuracy over speed

I. Practice Questions

1. If ZEBRA is coded as AFCSD, how is APPLE coded?
2. In a code, A=26, B=25, C=24... Z=1. What is BABY coded as?
3. LISTEN is coded as KRLNDO. How is SILENT coded?

Answers: 1. BQQMF 2. 2-1-2-25 3. RKIOBM

2 BLOOD RELATIONS

A. What is it?

Understanding family trees and relationships between people. You're given statements like "A is B's father" and need to figure out "Who is C to D?"

B. Core Concepts

Basic Relations: Father, Mother, Son, Daughter, Brother, Sister, Uncle, Aunt, Cousin, Nephew, Niece, Grandfather, Grandmother, Grandson, Granddaughter

By Marriage: Husband, Wife, Father-in-law, Mother-in-law, Brother-in-law, Sister-in-law, Son-in-law, Daughter-in-law

Gender Neutral Terms: Parent, Sibling, Child, Ancestor, Descendant

C. Types of Questions

- Direct relationship (A is B's what?)
- Indirect relationship (Through chain of relations)
- Coded relationships ($A + B$ means A is father of B)
- Arrangement-based (Draw family tree and answer)

D. Step-by-Step Solving Approach

1. **Draw family tree:** Visual representation helps immensely
2. **Mark generations:** Use different lines for descent/siblings
3. **Trace path:** From person A to person B through the tree
4. **Identify relationship:** Based on the path you traced
5. **Remember gender:** Some relations differ by gender (uncle vs aunt)

E. Shortcuts & Tricks

💡 **Gender Elimination:** If answer asks "she", eliminate uncle, brother, etc.

💡 **Generation Check:** Same generation = sibling/cousin; one down = child/nephew; one up = parent/uncle

💡 **Reverse Method:** Instead of $A \rightarrow B$, think $B \rightarrow A$. Sometimes easier!

💡 **Coded Relations Quick Rules:** $A+B$ (A father of B), $A-B$ (A mother of B), $A \times B$ (A sibling of B), $A \div B$ (A spouse of B)

F. Solved Examples

BASIC: A is B's father. B is C's mother. What is A to C?

Solution:

$A \rightarrow \text{Father} \rightarrow B \rightarrow \text{Mother} \rightarrow C$

A is one generation above B, B is one generation above C

Therefore, A is two generations above C

Answer: **Grandfather (Paternal)**

INTERMEDIATE: P is Q's brother. Q is R's mother. S is R's father. What is P to S?

Solution:

$P (\text{male}) \rightarrow \text{Brother} \rightarrow Q \rightarrow \text{Mother} \rightarrow R$

$S \rightarrow \text{Father} \rightarrow R$

So: S is R's father, Q is R's mother

P and Q are siblings

S is married to Q

Therefore, P is S's **Brother-in-law**

ADVANCED: If $A+B$ means A is father of B, $A-B$ means A is mother of B, $A \times B$ means A is sister of B, $A \div B$ means A is brother of B. What relation is P to T in expression: $P+Q-R \times S \div T$?

Solution:

$P+Q$: P is father of Q

$Q-R$: Q is mother of R

$R \times S$: R is sister of S

$S \div T$: S is brother of T

Building family:

$P \rightarrow \text{Father} \rightarrow Q \rightarrow \text{Mother} \rightarrow R = S (\text{same person}) \rightarrow \text{Brother} \rightarrow T$

P and Q are spouses (P husband, Q wife)

R and S are siblings (R is female, S is male)

S and T are siblings (S is male, T is younger)

So P is R's father, and R is S (same).

P is T's **Father**

G. Common Mistakes

✗ Not drawing the family tree and getting confused

✗ Confusing paternal and maternal relations

✗ Forgetting that "she/he" gives gender clues

H. Exam Strategy

- Always draw family tree, even if it takes 20 seconds. Prevents errors
- When confused, try reverse direction (B to A instead of A to B)
- Time: 1-2 min per question in placements, 2-3 min in CAT

I. Practice Questions

1. Introducing a woman, Hari said "She is the mother of the only child of my mother." Who is the woman to Hari?
2. A and B are married. A is C's father. B is D's sister. What is C to D?
3. If P is Q's brother, Q is R's mother, then P is R's?

Answers: 1. Wife 2. Cousin (male to female) 3. Uncle

3 DIRECTION SENSE

A. What is it?

Someone walks/moves in different directions. You need to figure out their final position relative to starting point. It's like tracking someone's movement on a map.

B. Core Concepts

Basic Directions: North, South, East, West, Northeast, Northwest, Southeast, Southwest

Left & Right: Depend on person's current facing direction

Distance: Can be given in units, km, meters, or steps

Turns: 90°, 180°, etc. Or "left turn", "right turn"

C. Types of Questions

- Final position relative to starting point
- Distance from starting point
- Direction from starting point
- Number of turns made

D. Step-by-Step Solving Approach

1. **Draw compass:** N at top, S at bottom, E at right, W at left
2. **Mark starting point:** Usually assume person starts at origin facing North
3. **Trace each movement:** Update position and facing direction after each move
4. **Calculate final position:** Use Pythagoras if needed for distance
5. **Determine direction:** From starting point to final point

E. Shortcuts & Tricks

💡 **Use Coordinate System:** North = +Y, South = -Y, East = +X, West = -X

💡 **Track Only Final Coordinates:** Don't worry about path, just where you end up

💡 **Left/Right from Person's View:** If facing North, left = West, right = East

💡 **Pythagorean Theorem:** If you move X units horizontally and Y units vertically, distance = $\sqrt{X^2 + Y^2}$

F. Solved Examples

BASIC: Person starts at point A, walks 10m North, then 5m East, then 10m South. How far and in which direction from starting point?

Solution:

Start: (0,0)

After 10m North: (0,10)

After 5m East: (5,10)

After 10m South: (5,0)

Final position: 5m East of starting point

Answer: **5m East**

INTERMEDIATE: Person faces North. Turns 90° right, walks 20m. Turns 90° left, walks 15m. Turns 90° left, walks 20m. What is distance and direction from start?

Solution:

Start: Facing North at (0,0)

90° right: Now facing East

Walk 20m: At (20,0), still facing East

90° left: Now facing North

Walk 15m: At (20,15), still facing North

90° left: Now facing West

Walk 20m: At (0,15), facing West

Final position: 15m North of starting point

Answer: **15m North**

ADVANCED: Person walks 12m West, then turns 90° clockwise and walks 5m, turns 90° counterclockwise and walks 13m. Find shortest distance to starting point.

Solution:

Start: (0,0) facing North

12m West: (-12,0)

Turn 90° clockwise (facing West) → now facing North (error in logic)

Actually: If facing North, turn 90° clockwise = facing East

Walk 5m East: (-12,0) → (-7,0)

Turn 90° counterclockwise (facing East) → now facing North

Walk 13m North: (-7,13)

Distance = $\sqrt{((-7)^2 + 13^2)} = \sqrt{(49 + 169)} = \sqrt{218} \approx 14.76\text{m}$

G. Common Mistakes

✗ Confusing left and right based on person's perspective, not yours

✗ Forgetting to update facing direction after turns

✗ Clockwise vs counterclockwise confusion

H. Exam Strategy

- Always draw a simple diagram. It takes 30 seconds but prevents errors
- Use coordinate system for accuracy
- Time: 1-2 min per question

I. Practice Questions

1. Ravi walks 8km East, turns left, walks 6km. What distance from starting point?
2. Facing North, turn 90° left, walk 100m. Turn 90° right, walk 50m. Direction and distance?

Answers: 1. 10km (8 East, 6 North, Distance= $\sqrt{64+36}=10$) 2. 100m North, 50m East from start = $\sqrt{100^2+50^2}\approx 111.8\text{m}$, NE direction

4 SEATING ARRANGEMENT

A. What is it?

People sit in a specific pattern. You're given clues about their positions and need to figure out the complete arrangement. Like solving a puzzle to place people in seats.

B. Core Concepts

Linear Arrangement: People sit in a straight line. Left and right are clear.

Circular Arrangement: People sit in a circle. No absolute left/right; relative to direction of face matters.

Rectangular/Square Table: Multiple sides. Need to understand sides and positions.

C. Types of Questions

- Find person at specific position
- Find position of specific person

- Who sits between two people?
- Who sits to left/right of a person?

D. Step-by-Step Solving Approach

1. **Draw the seating arrangement:** Line or circle based on problem
2. **Number positions:** 1-n from left to right (linear) or any starting point (circular)
3. **List all clues:** Break complex statements into simple facts
4. **Apply clues one by one:** Some clues directly place people; use them first
5. **Use elimination:** If A is not position 1,2,3... only position 4,5... remains
6. **Verify with all clues:** Make sure final arrangement satisfies every clue

E. Shortcuts & Tricks

- 💡 **Start with Fixed Clues:** "A sits at left end" or "B is 3rd from left" → place them first
- 💡 **Use Elimination:** If 5 people and 4 positions are known, 5th person goes to remaining position
- 💡 **Relative Position Clues:** "A sits to right of B" combined with other clues narrows down positions
- 💡 **In Circular:** Clockwise/counterclockwise matters. Right of A clockwise $\neq$ right of A counterclockwise
- 💡 **Create Multiple Scenarios:** If one clue has two possibilities, try both and see which satisfies all clues

F. Solved Examples

BASIC (LINEAR): A, B, C, D, E sit in a line. A is left of C. C is left of E. B is between A and C. Who sits 2nd?

Solution:

A left of C: A...C

C left of E: C...E

B between A and C: A-B-C

Combining: A-B-C...E

Need to place D. D can be after C and before E, or after E.

Check all clues: A-B-C-D-E or A-B-C-E-D both work... need more info

Actually, from clues given: A-B-C-(D,E in some order)

If no other clues: B sits 2nd or try A-B-C-D-E

Answer: **B sits 2nd**

INTERMEDIATE (CIRCULAR): A, B, C, D, E sit in circle. A is right of B. C is opposite D. A is not opposite E. B is right of D. Who is opposite A?

Solution:

In circle of 5, position opposite person at position n is person at position (n+2 or n+3 depending on counting)

Let's number clockwise: 1,2,3,4,5

Place A at position 1

A right of B (clockwise from B to A): B could be at 5

B right of D: If B=5, D right of B counterclockwise? Or D is such that B is right of D?

This becomes complex. Need to systematically try.

C opposite D: If C=1, D=3. If C=2, D=4, etc.

[Detailed placement would require trying combinations]

Solution requires systematic placement or trial.

G. Common Mistakes

- ✗ In circular, confusing "right" (clockwise) with "right" from person's face direction
- ✗ Not verifying answer against ALL clues
- ✗ Assuming positions without clear clues

H. Exam Strategy

- PLACEMENTS: Usually linear, 5-6 people, 4-5 straightforward clues. Time: 3-5 min
- CAT: Can be complex circular, 6-8 people, intricate clues. Time: 8-10 min for set
- Draw clearly and systematically. Don't guess.

I. Practice Questions

1. A, B, C, D sit in line. C is left of D. A is left of B. A is not leftmost. Who is leftmost?

Answers: 1. C (from clues: C-D and A-B arrangements, A not leftmost, so C-A-B-D or C-D-A-B. Leftmost is C)

5 PUZZLES (Box, Floor, Scheduling, Logic Grid)

A. What is it?

A puzzle is a complex arrangement problem where multiple people/items are placed in different locations/times with various conditions. You need to use all clues together to figure out the complete arrangement. It's like solving a detective mystery.

B. Core Concepts

- **Box Puzzle:** Items placed in boxes with conditions (e.g., "Red box is not next to blue box")
- **Floor Puzzle:** People on different floors (e.g., "A lives 2 floors above B")
- **Scheduling:** Events at different times with conditions
- **Logic Grid:** Cross-tabulation method to match attributes
- **Set-based Puzzle:** Groups of items with overlapping properties

C. Types of Questions

- Who/What is at position X?
- Which items/people are together?
- What is the arrangement?
- Multiple questions on same puzzle

D. Step-by-Step Solving Approach

1. **Identify puzzle type:** Box, floor, scheduling, or logic grid?
2. **List all variables:** People, positions/floors, items, times, properties
3. **Create grid/table:** Visual representation of all possibilities
4. **List all clues:** Break complex statements into simple facts
5. **Apply direct clues first:** Clues that directly place items (eliminates possibilities)
6. **Use elimination:** If A can't be 1,2,3, then A must be 4
7. **Apply indirect clues:** Use relationships between placed items

8. **Verify all clues:** Make sure solution satisfies EVERY clue

E. Shortcuts & Tricks

- 💡 **Create Logic Grid:** Rows = one variable, Columns = another. Mark X for impossible, checkmark for possible.
- 💡 **Direct Clues First:** "A is 1st" → Mark A at position 1. This eliminates other positions for A.
- 💡 **Cascading Elimination:** If A is determined, and "B is not next to A", eliminate those positions for B.
- 💡 **Look for Unique Solutions:** If only one position remains, that item **MUST** go there.
- 💡 **Contradiction Method:** If assuming position leads to contradiction, that position is wrong.

F. Solved Examples

BASIC: 3 boxes (Red, Blue, Green) with items (Apple, Ball, Car). Red box doesn't have Apple. Blue box doesn't have Ball. Green box doesn't have Car. What's in each box?

Solution:

Clue 1: Red $\neq$ Apple → Red has Ball or Car

Clue 2: Blue $\neq$ Ball → Blue has Apple or Car

Clue 3: Green $\neq$ Car → Green has Apple or Ball

Test: If Red has Apple - violates clue 1. So Red has Ball or Car.

If Red has Ball: Blue has Apple or Car, Green has Apple or Ball

Since Red has Ball, Green must have Apple (from clue 3)

Then Blue has Car

Answer: **Red=Ball, Blue=Car, Green=Apple**

INTERMEDIATE: 5 people (A,B,C,D,E) on 5 floors. A is below D. C is above B. D is not on floor 1. E is on floor 5. Where is B?

Solution:

E is on floor 5 (given)

D is not on floor 1, so D is on 2,3,4 (not 5 because E is there)

A is below D, so A must be lower than D

C is above B

If D=4, A can be 1,2,3

If D=3, A can be 1,2

If D=2, A must be 1

Since C above B, and we have 5 people: 1,2,3,4,5

One solution: A=1, B=2, C=3, D=4, E=5

Check: A(1) below D(4)✓, C(3) above B(2)✓, D not on 1✓, E on 5✓

Answer: **B is on floor 2** (though other solutions possible)

ADVANCED: Logic grid puzzle with 4 people (Jack, Kate, Lee, Max), 4 jobs (Doctor, Lawyer, Pilot, Chef), 4 cities (Paris, London, Rome, Berlin). Doctor is in Paris. Jack is not Doctor. Kate is not Lawyer. If Lee is Pilot, then Lee is in Rome. Max is not in London. Where is Kate and what is her job?

Solution Method:

Create 4x4 grid for people vs jobs, 4x4 for jobs vs cities

Direct clues:

- Doctor in Paris (crosses out other cities for Doctor job)
- Jack ≠ Doctor
- Kate ≠ Lawyer
- Lee = Pilot → Lee in Rome
- Max ≠ London

Assume Lee = Pilot: Then Lee in Rome

Remaining: Jack, Kate, Max for Doctor, Lawyer, Chef

Jack ≠ Doctor, Kate ≠ Lawyer

So: Either Jack=Lawyer/Chef, Kate=Doctor/Chef, Max=Doctor/Lawyer/Chef

If Kate = Doctor: Kate in Paris

Then Jack and Max are Lawyer and Chef

Jack can be Lawyer or Chef

Max ≠ London, so Max in Berlin (since Paris=Kate, Rome=Lee)

Jack in London

One solution: **Kate = Doctor in Paris**

G. Common Mistakes

✗ Assuming a solution without verifying all clues

- ✗ Not using elimination properly (crossing out possibilities)
- ✗ Forgetting that each item/person has exactly one position
- ✗ Not creating visual grid/table (working in head leads to errors)

H. Exam Strategy

- PLACEMENTS: Usually 3-4 items with 5-6 clues. Time: 5-7 min per puzzle
- CAT: Complex puzzles, 5-7 items, intricate clues. Time: 10-15 min for entire set
- Always draw a grid/table. Never rely on mental calculation for puzzles.

I. Practice Questions

1. 4 people (P,Q,R,S) have 4 colors (Red, Blue, Green, Yellow). P doesn't have Red. Q doesn't have Blue. R has Yellow or Green. S has Green or Red. If R has Yellow, where does Q have? (Try to work through this using elimination)

6 SYLLOGISM

A. What is it?

Two statements are given (premises) and you need to determine what must logically follow (conclusion). It's about finding what **MUST** be true based on the given information.

B. Core Concepts

All A are B: Every member of A is also in B. Represented as: $A \subseteq B$

No A are B: Complete separation. A and B have no overlap.

Some A are B: At least one member of A is in B. Could be partial overlap or complete overlap.

Some A are not B: At least one member of A is outside B.

C. Venn Diagram Approach

All A are B: B [A inside B] | B | | | A | | | | No A are B: A B
(completely separate, no overlap) Some A are B: A B | A only | AB | B only |

D. Step-by-Step Solving Approach

1. **Identify statement type:** All, No, Some, Some...not
2. **Draw Venn diagram:** Visual representation of each statement
3. **Identify subject and predicate:** Which is A, which is B
4. **Find the middle term:** Common element in both statements
5. **Determine conclusion:** What must be true given both premises
6. **Check against all possibilities:** Can conclusion be false? If yes, it's not valid

E. Shortcuts & Tricks

💡 **"All + All" Rule:** All A are B + All B are C $\rightarrow$ All A are C

💡 **"No" eliminates:** "No A are B" means zero intersection. Complete separation.

💡 **"Some" is weak:** "Some A are B" doesn't tell us about "Some A are not B" (both can be true)

💡 **Negative rule:** If one premise is negative, conclusion must be negative

💡 **Two "Some" = No valid conclusion:** "Some A are B" + "Some B are C" doesn't determine if A and C overlap

F. Solved Examples

BASIC: All cats are animals. Momo is a cat. Is Momo an animal?

Solution:

Statement 1: All cats are animals

Statement 2: Momo is a cat

Venn diagram: Cats are inside Animals. Momo is a cat, so Momo is inside Animals.

Conclusion: **Yes, Momo is an animal (Must be true)**

INTERMEDIATE: All students are hardworking. Some hardworking people are rich. Are all students rich?

Solution:

Statement 1: All students are hardworking

Statement 2: Some hardworking people are rich

Venn diagram:

- Students circle is completely inside Hardworking circle
- Hardworking circle partially overlaps with Rich circle

Students could be in the hardworking part that overlaps with Rich,
OR in the hardworking part that doesn't overlap with Rich.

Conclusion: **Cannot determine. Not necessarily all students are rich.**

ADVANCED: No flowers are animals. Some plants are flowers. Are no plants animals?

Solution:

Statement 1: No flowers are animals (Flowers and Animals completely separate)

Statement 2: Some plants are flowers

Venn diagram:

- Flowers and Animals don't overlap
- Plants partially overlap with Flowers
- Plants could also overlap with Animals (not flowers part)

Some plants are flowers (those don't overlap with animals)

But some plants might not be flowers, and those could be animals

Conclusion: **Cannot say "No plants are animals." Some plants might be animals.**

G. Common Mistakes

✗ Assuming "Some A are B" means "Some A are not B" (not necessarily true)

✗ Not using Venn diagram (mental logic errors common)

✗ Reversing statements (All A are B $\neq$ All B are A)

H. Exam Strategy

- Always draw Venn diagram. Visual thinking eliminates mistakes.
- Time: 1-2 min per question (if using diagram correctly)
- Three possible answers: Must be true, Must be false, Cannot be determined

I. Practice Questions

1. All birds have wings. Penguins are birds. Do penguins have wings?
2. Some doctors are rich. All rich people are happy. Are all doctors happy?
3. No lies are truths. Some statements are lies. Are some statements not truths?

Answers: 1. Yes, must be true 2. Cannot determine 3. Yes, must be true

7 STATEMENT & CONCLUSION

A. What is it?

You're given a passage/statement and then several conclusions. You need to determine which conclusions logically follow from the statement. It's about reading carefully and deducing only what can be proven from given information.

B. Core Concepts

Must Be True: Conclusion that MUST follow from statement. No other possibility.

Can Be True: Conclusion that could be true, but isn't necessarily. Example: "Some students failed" doesn't mean "All students failed" but that's possible.

Cannot Be True: Conclusion that contradicts the statement or is impossible.

C. Types of Questions

- Which conclusion follows?
- Which conclusion must be true?
- Which conclusion can be inferred?
- How many conclusions follow?

D. Step-by-Step Solving Approach

1. **Read statement carefully:** Note exact words (all, some, no, none)
2. **Extract main claims:** What is the statement saying?
3. **For each conclusion, ask:** "Must this be true given the statement?"
4. **If it COULD be false:** It's not a valid conclusion
5. **Mark each conclusion:** Must follow, Can follow, Cannot follow

E. Shortcuts & Tricks

💡 **"All" is strongest:** "All X are Y" means every X is Y. Conclusion about all X must be true.

💡 **"Some" is weak:** "Some X are Y" doesn't tell us about the rest. Can't conclude "All X" or "No X".

💡 **Look for absolute words:** Always, Never, All, None are definite. Must be true statements are based on these.

💡 **Exceptions rule:** "All except" means there's one exception. Read carefully.

F. Solved Examples

BASIC: "All students who passed the exam are talented." Conclusion: "All talented people passed the exam."

Solution:

Statement says: Passed exam → Talented

Conclusion says: Talented → Passed exam

This is a reversal! Not valid.

Example: A talented person might not have passed the exam.

Answer: **Conclusion does NOT follow**

INTERMEDIATE: "Some politicians are corrupt. All corrupt people should be removed from office." Conclusion: "Some politicians should be removed from office."

Solution:

Premise 1: Some politicians are corrupt

Premise 2: All corrupt people → should be removed

Linking: Some politicians (the corrupt ones) are corrupt (premise 1)

Those corrupt ones should be removed (premise 2)

Answer: **Yes, conclusion MUST follow**

G. Common Mistakes

- ✗ Reversing statements (All A are B $\neq$ All B are A)
- ✗ Overgeneralizing from "some"
- ✗ Bringing in outside knowledge instead of using only given information

H. Exam Strategy

- Write down exactly what the statement says, don't interpret
- For each conclusion, ask: "Can this be false?"
- Time: 1-2 min per question

I. Practice Questions

1. "All successful people work hard." Conclusion: "All hardworking people are successful."
2. "No fool is wise." Conclusion: "Some wise people are not fools."

Answers: 1. Does NOT follow (reversal) 2. MUST follow

A. What is it?

An assumption is an unstated belief that the speaker assumes to be true for their statement to make sense. You need to find what **MUST** be assumed for the statement to be valid.

B. Core Concepts

Explicit (Stated): Information directly given in the statement.

Implicit (Assumed): Information not stated but necessary for statement to make sense.

Reasonable Assumption: Logically necessary for statement to be true.

Unreasonable Assumption: Not necessary; statement can be true without it.

C. Step-by-Step Solving Approach

1. **Understand the statement:** What is being claimed?
2. **Ask "Why?":** Why would the speaker say this?
3. **Find missing links:** What unstated information is needed?
4. **Test assumption:** If this assumption is false, does statement still work?
5. **If "No":** It's a valid assumption

E. Shortcuts & Tricks

💡 **Method of Denial:** If assuming opposite makes statement false, it's a valid assumption.

💡 **Cause-Effect assumptions:** "Due to X, Y happened" assumes X causes Y.

💡 **Comparison assumptions:** "A is better than B" assumes we can compare A and B.

F. Solved Examples

BASIC: "Don't use plastic bags; they harm the environment." What is assumed?

Solution:

Statement: Don't use plastic bags because they harm environment

Assumption: Harming the environment is bad / We should avoid things that harm environment

If we didn't assume environment is valuable, the statement makes no sense.

Answer: **Assumes environment is important/valuable**

INTERMEDIATE: "Since Sita is a woman, she will be a good mother." What is assumed?

Solution:

Statement claims: Being woman → will be good mother

This assumes: Women are naturally suited to motherhood

Or: All women will be good mothers

If we didn't assume women are inherently good at mothering, the statement is unfounded.

Answer: **Assumes women are naturally good mothers**

G. Common Mistakes

- ✗ Confusing implications with assumptions
- ✗ Assuming something that doesn't need to be assumed

H. Exam Strategy

- Use denial test: Assume opposite and see if statement breaks
- Time: 1-2 min per question

I. Practice Questions

1. "Students who study hard will pass the exam." What is assumed?

Answer: Assumes studying hard leads to passing / Hard work correlates with exam success